

Magnetic Effects of Electric Current

Magnetic field and field lines, field due to current in a wire, loop and solenoid, force on a current carrying conductor, Fleming rules, electric motor, generator and household wiring.

Oersted Experiment

In 1820, Hans Christian Oersted discovered that electricity and magnetism are connected. He showed that a current carrying wire produces a magnetic field around it.

Experiment: A magnetic compass is placed near a wire. When current flows through the wire, the compass needle deflects. When current is switched off, the needle returns to its original position. When direction of current is reversed, the needle deflects in the opposite direction.

Conclusion: A current carrying conductor behaves like a magnet and produces a magnetic field around it. The direction of the field depends on the direction of current. This is called the magnetic effect of current.

Magnet and Magnetic Field

Magnet

A magnet attracts iron, cobalt and nickel. Every magnet has two poles, North and South. Like poles repel and unlike poles attract. A magnet exerts a force on another magnet without contact, through its field.

Magnetic field

Magnetic field is the region around a magnet or a current carrying conductor where its magnetic force can be felt. It is a vector quantity. Its SI unit is tesla, denoted by T.

A magnetic compass needle is a small bar magnet that shows the direction of the field. The north pole of the compass points in the direction of the magnetic field at that point.

Magnetic Field Lines

Magnetic field lines are imaginary lines that show the pattern of the magnetic field. They are also called magnetic lines of force.

Properties of magnetic field lines:

- They are closed continuous curves. Outside the magnet they go from North pole to South pole, and inside the magnet from South pole to North pole.
- The tangent at any point on a field line gives the direction of the magnetic field at that point.
- No two field lines intersect. If they did, it would mean two directions of the field at the same point, which is not possible.
- Field lines are crowded where the field is strong and spread out where the field is weak.
- Lines are closer near the poles and farther away from the magnet.

Pattern of field lines for a bar magnet can be seen by sprinkling iron filings around the magnet. The filings arrange along the field lines.

Magnetic Field due to Current

A current carrying conductor creates a magnetic field around it. The shape of the field depends on the shape of the conductor.

Straight current carrying conductor

The field lines around a straight wire are concentric circles centred on the wire. The plane of the circles is perpendicular to the wire.

Direction is given by the Right Hand Thumb Rule:

Right Hand Thumb Rule

Imagine holding the current carrying conductor in your right hand, with the thumb pointing in the direction of current. Then the fingers curl around the conductor in the direction of the magnetic field lines.

Magnitude: The field is stronger close to the wire and weaker farther away. It is directly proportional to current and inversely proportional to distance from the wire.

Circular loop

The field lines due to a circular loop are circular near the wire but become straighter near the centre. At the centre, the field lines are almost straight and perpendicular to the plane of the loop. The field at the centre is strongest and its direction is perpendicular to the plane.

Factors: Magnetic field at the centre is directly proportional to current and inversely proportional to radius of the loop. If there are n turns in the coil, the field is n times larger. So a circular coil with many turns produces a stronger field.

Clock face rule: If current appears clockwise when seen from one face, that face is South pole. If current appears anticlockwise, that face is North pole.

Solenoid

A solenoid is a long coil of many turns of insulated copper wire wound closely in the shape of a cylinder. When current flows, it behaves like a bar magnet.

- The field lines inside a solenoid are parallel straight lines. This shows the field inside is uniform.
- One end acts as North pole and the other as South pole, depending on direction of current.
- The field inside a solenoid depends on current and number of turns per unit length. More current or more turns gives stronger field.
- A soft iron core inside the solenoid increases the field and makes an electromagnet.

Difference: A bar magnet is a permanent magnet. A solenoid is an electromagnet that works only when current flows, and its strength can be changed.

Electromagnet and Permanent Magnet

An electromagnet is a temporary magnet made by winding a coil around a soft iron core and passing current. It is strong but loses magnetism when current is off. A permanent magnet is made from hard steel, keeps its magnetism and cannot be easily demagnetised.

Force on a Current Carrying Conductor

A current carrying conductor placed in a magnetic field experiences a force. This was shown by experiments where a small aluminium rod on rails moves when current is passed through it inside a magnetic field.

Direction of force is perpendicular to both the direction of current and the direction of magnetic field.

Fleming's Left Hand Rule (Motor Rule)

Stretch the thumb, forefinger and middle finger of your left hand so that they are perpendicular to each other. If the forefinger points in the direction of magnetic field and the middle finger in the direction of current, then the thumb points in the direction of motion or force on the conductor.

Use: Current is the flow of charge, and moving charges in a magnetic field feel a force. The same rule applies. The magnitude of force is maximum when current is perpendicular to the field and zero when current is parallel to the field.

Electric Motor

CBSE 2026-27 assessment note: the electric motor, electromagnetic induction and electric generator are prescribed for formative assessment only and are not part of the year-end board examination.

An electric motor is a device that converts electrical energy into mechanical energy. It works on the principle that a current carrying coil in a magnetic field experiences a force and rotates.

Principle

When a rectangular coil carrying current is placed in a magnetic field, opposite sides of the coil experience forces in opposite directions. These two forces form a couple that rotates the coil.

Construction

- **Armature coil:** A rectangular coil of insulated copper wire wound on a soft iron core.
- **Magnet:** Two pole pieces of a magnet, North and South, that provide a strong magnetic field.
- **Split ring commutator:** Two half rings of copper, called R1 and R2, connected to the two ends of the coil. It reverses the direction of current in the coil after every half rotation.
- **Brushes:** Two carbon brushes that press against the commutator and connect the coil to the battery.
- **Battery:** Source of current.

Working

Current enters the coil through one brush and one half ring, flows through the coil and leaves through the other half ring and brush. The magnetic field exerts opposite forces on the two sides of the coil and the coil rotates. After half a turn, the commutator reverses the contact, so current in the coil reverses, but the forces continue to rotate the coil in the same direction. So the coil keeps rotating continuously.

Function of split ring: To reverse current after every half rotation so that the coil keeps rotating in the same direction. Without it, the coil would oscillate and stop.

Function of brushes: To make contact between the rotating coil and the fixed circuit.

Speed of motor can be increased by increasing current, number of turns in the coil, strength of magnetic field and area of the coil. Commercial motors use an electromagnet instead of a permanent magnet, many turns on a soft iron core, and a more efficient commutator.

Electromagnetic Induction

Michael Faraday discovered the opposite effect in 1831. A changing magnetic field can produce electric current. This is called electromagnetic induction.

Experiment: A coil is connected to a galvanometer. When a magnet is moved towards the coil, the galvanometer shows a deflection. When the magnet is moved away, the deflection is in the opposite direction. When there is no relative motion, there is no current. Faster motion gives larger deflection.

Current can also be produced by moving the coil towards a fixed magnet, or by changing the current in a nearby coil. The induced current is larger when the motion is faster, the magnet is stronger or the coil has more turns.

Fleming's Right Hand Rule (Generator Rule)

Fleming's Right Hand Rule

Stretch the thumb, forefinger and middle finger of your right hand so that they are perpendicular to each other. If the forefinger points in the direction of magnetic field and the thumb in the direction of motion of the conductor, then the middle finger points in the direction of induced current.

Electric Generator

An electric generator converts mechanical energy into electrical energy. It works on the principle of electromagnetic induction. When a coil is rotated in a magnetic field, the changing magnetic flux through the coil induces a current.

AC Generator

- **Construction:** Rectangular coil ABCD, strong field magnets, two slip rings R1 and R2, two brushes B1 and B2.
- **Working:** When the coil is rotated, the arm AB moves up and CD moves down. Using Fleming's right hand rule, current flows from A to B and C to D. After half rotation, AB moves down and CD moves up, so current reverses. So the current changes direction after every half rotation. This is alternating current, AC.

- **Slip rings:** Each end of the coil is connected to a full ring, not a split ring. Brushes maintain contact as the coil rotates. They allow the induced AC to go out without reversing inside the coil.

DC Generator

A DC generator is similar to an AC generator, but it uses a split ring commutator instead of two slip rings. The commutator reverses the connection after every half turn, so the current in the external circuit flows in the same direction. This is direct current, DC.

Difference between AC and DC

AC changes direction periodically and is used for household supply because it can be transmitted over long distances with less loss using transformers. DC flows in one direction only and is used for batteries, cells and electronic devices.

Domestic Electric Circuits

Electricity is supplied to homes through a main supply from a pole. It has two main wires, live and neutral, and an earth wire for safety.

- **Live wire:** Carries the alternating supply potential to the appliance.
- **Neutral wire:** Completes the circuit and is maintained near earth potential.
- **Earth wire:** Provides a low-resistance path from the metal body of an appliance to earth, reducing electric-shock risk if insulation fails. The NCERT diagram uses red, black and green; modern installations may follow newer colour codes.

Potential difference between live and neutral is 220 V, frequency is 50 Hz in India.

Wiring: All appliances are connected in parallel, so each gets 220 V. Each appliance has a separate switch connected on the live side. A main fuse and an earth leakage circuit breaker are connected at the meter board.

Fuse

A fuse is a safety device made of a low melting point alloy. It is connected in series with the live wire. If current exceeds a safe value due to overloading or short circuit, the fuse wire heats and melts, breaking the circuit and preventing fire. Now MCBs, miniature circuit breakers, are also used.

Overloading and Short Circuit

- **Overloading:** Drawing too much current by connecting too many appliances on one line or using a high power appliance. It heats the wires and can cause fire.
- **Short circuit:** When live and neutral wires touch directly, resistance becomes almost zero and a very large current flows. This produces a lot of heat and can cause fire. Fuse prevents this.

Summary

- Oersted showed that current produces a magnetic field. Field lines are closed curves, do not intersect and show direction and strength of field.
- Field around a straight wire is in concentric circles. Direction by right hand thumb rule. Field of a circular loop and a solenoid depends on current and number of turns. Solenoid acts like a bar magnet.
- A current-carrying conductor in a magnetic field feels a force. Its direction is given by Fleming's left-hand rule; this is the principle of the electric motor.
- Electric motor converts electrical energy to mechanical energy. Split ring commutator reverses current after every half turn to keep rotation continuous.
- Electromagnetic induction is production of current by changing magnetic field. Direction by Fleming's right hand rule.
- Generator converts mechanical energy to electrical energy. AC generator uses slip rings and gives AC. DC generator uses split ring commutator and gives DC.
- Household supply is 220 V, 50 Hz, live and neutral in parallel, earth wire for safety, fuse or MCB to prevent overloading and short circuit.

Practice Questions

1. Describe the Oersted experiment and state the conclusion. What happens when the direction of current is reversed?

2. State the properties of magnetic field lines. Why do they not intersect? Draw the pattern for a bar magnet.

3. State the right hand thumb rule. Draw the magnetic field pattern around a straight wire and a circular loop. How does a solenoid behave like a magnet?

4. State Fleming's left hand rule. Explain the principle, construction and working of an electric motor. What is the role of the split ring commutator?

5. What is electromagnetic induction? Explain with an activity. State Fleming's right hand rule. Distinguish between AC and DC generator.

6. Explain domestic wiring in India: colours of live, neutral and earth wire, why appliances are in parallel, and the function of earth wire and fuse. What are overloading and short circuit?